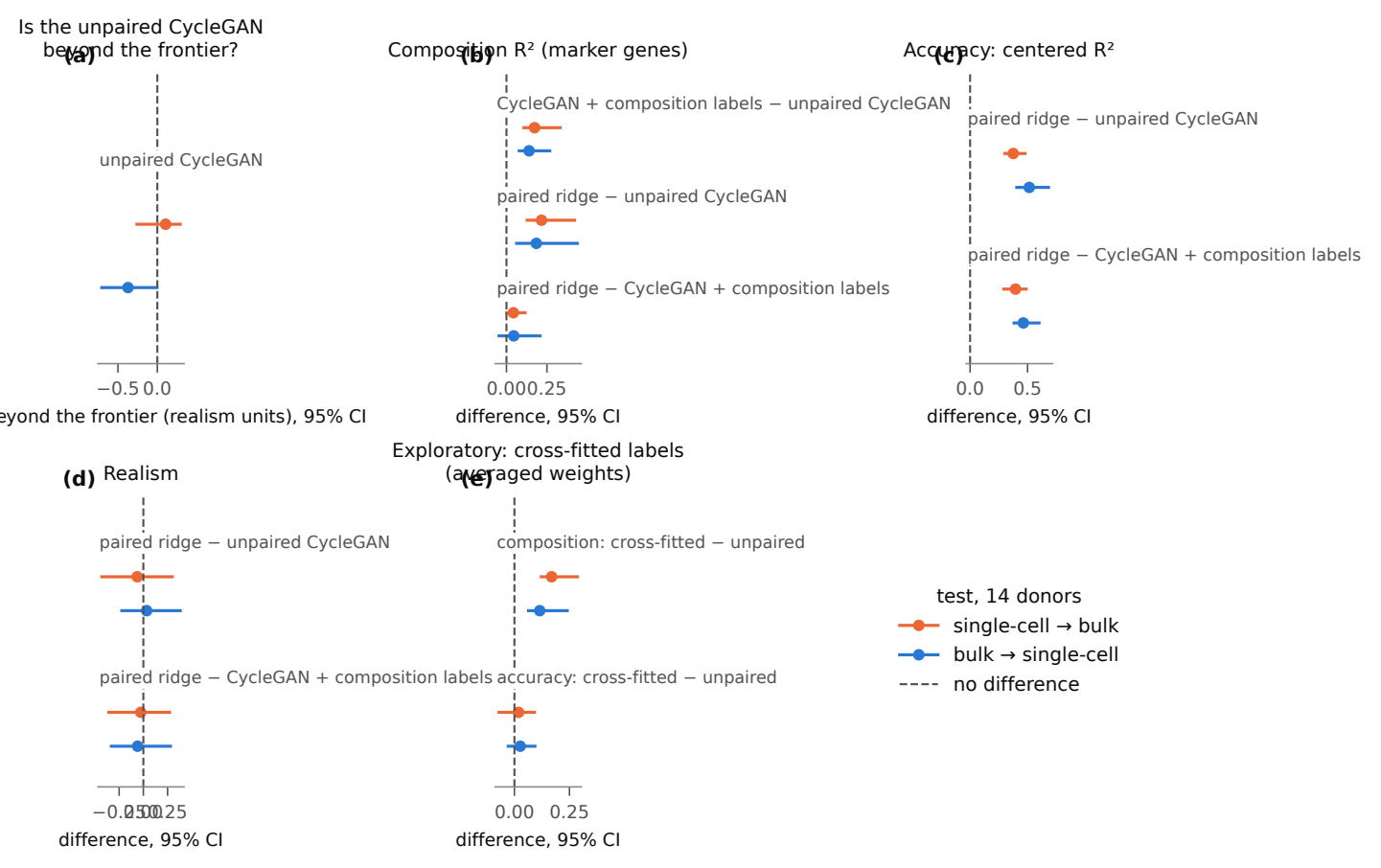
Is the unpaired CycleGAN beyond the frontier?
(a)
unpaired CycleGAN
−0.5 0.0
eyond the frontier (realism units), 95% CI
Composition R² (marker genes)
(b)
CycleGAN + composition labels − unpaired CycleGAN
paired ridge − unpaired CycleGAN
paired ridge − CycleGAN + composition labels
0.000.25
difference, 95% CI
Accuracy: centered R²
(c)
paired ridge − unpaired CycleGAN
paired ridge − CycleGAN + composition labels
0.0 0.5
difference, 95% CI
(d) Realism
paired ridge − unpaired CycleGAN
paired ridge − CycleGAN + composition labels
−0.250.00.25
difference, 95% CI
Exploratory: cross-fitted labels (averaged weights)
(e)
composition: cross-fitted − unpaired
accuracy: cross-fitted − unpaired
0.00 0.25
difference, 95% CI
test, 14 donors
single-cell → bulk
bulk → single-cell
no difference